INDEX

Veera's Student's Choice

Name : S. MUTHU KRISHNAN

STD : ..... SEC : ..... ROLL NO : 192411168

S.No.	Date	Title	~~Page No.~~ Mark	Teacher's Sign /
1	26.2.2026	Ceaser cipher	9	✓
2	26.2.2026	Monoalphabetic cipher	9	✓
3	27.2.2026	vigenere cipher	8	✓
4	27.2.2026	transposition cipher	9	✓
5	28.2.2026	Hill cipher	9	✓
6	1.3.2026	DES algorithm	10	✓
7	1.3.2026	Deffie-Heilman algorithm	10	✓
8	2.3.2026	RSA algorithm	9	✓
9	2.3.2026	Triple DES algorithm	9	✓
10	3.3.2026	Blow fish algorithm	9	✓
11	3.3.2026	RC5 algorithm	9	✓

1. Convert plain text into cipher text.

Aim:

To convert plain text into cipher text.

Algorithm:

* Take plain text and an shift value.
* For each character in the text
  → It shifts a letter
  - Convert to number (0 - 25)
  - Add shift (Key value)
  - Apply modulo 26.
  - Convert back to letter.

Input:

Text = Hello
Key = 4
Output = LIPPS

Result

1. Convert plain text to monoalphabetic cipher text

Aim:

To convert plain text to monoalphabetic cipher text.

Algorithm:

1. Input
   - plain text
2. Initialize
   - create an empty string result.
3. For each character in plain text.
   - *. If character is an upper case letter.
   - *. Lookup if substitution is mapping.
   - *. Append the mapped letter to result.
   - *. Convert it into upper case. Find it substitution then convert back to lower case.
   - *. Append to result.
   - *. If character is no letter.

Input:

FRUIT - Input

YKXOZ - output

3 Convert plain text to vignere cipher text

Aim

To Convert plain text to vignere cipher text.

Algorithm:

1. Input
   * Read plain text
   * Read keyword
2. preparation
   * Convert both plain text to same case
   * Repeat the key ontill it matches the plain text.
3. Encryption process

   For each character in plain text
   * Convert plain text to number (0 to 25)
   * convert key letter to number
   * Add together with modulo 26.
   * Convert result back to letter.
   * If it is not a letter keep it unchanged.

Output.

The resulting string is a cipher text

Input: Geeks for geeks

Output: GCYCZFMLYLIM

4. Convert plain text into transposition cipher text

Aim:

To convert plain text into transposition cipher text (Railfence method)

Algorithm:

1. Read the text.
2. Remove all space from the plain text.
3. Write how many key value needed.
4. Draw the table with column equal to key.
5. Now arrange the plain text in zigzag method.
6. Read the text in the row wise.

output:

Combine letters get cipher text

Input : Apple is a nourished fruit.

output : Apesnuishri | pliaorsefut

Result

5 Convert the plain text into hill cipher text.

Aim:

To convert plain text into hill cipher text.

Algorithm:

1. Take the key value 2x2 matrix.
2. Enter the plain text and remove spaces.
3. If the plaintext is odd add 'x' value.
4. Write the numeric value of alphabetic.
5. Enter the key value you needed.
6. Enter plain text you want to convert into cipher text.

output:

cipher text shown in the display.

Input: $K_1 = 9$ plain text = FARM

$K_2 = 0$

$K_3 = 2$

$K_4 = 10$

Output: TKXY

6. To convert plain text to DES algorithm

Aim:

To implement DES algorithm to encrypt & decrypt a message using a secret key.

Algorithm:

1. Enter plaintext
2. Enter secret key.
3. convert plaintext to binary.
4. Apply permutation.
5. Split into left and right halves.
6. perform 16 rounds and encryption using round fiction.
7. Combine halves and apply final permutation.
8. Display encrypted binary output.
9. Lower process to decrypt.
10. Display original plaintext.

Output:

plaintext - apple

output - 0010011100100010101111111110111000001000000
10100110110110101001 0.

Decrypted text - apple

7 Convert plain text to Deffie-Hellman.

Aim:

TO implement Deffie-Hellman Key exchang to generate a shared secret key between two users.

Algorithm:

1. Enter a prime number P
2. Enter a generate g.
3. Alice chooses private key a.
4. Bob chooses private key b
5. Alice computer's $A = (g^a \bmod P)$
6. Bob computer's $B = (g^b \bmod P)$
7. Exchange A and B.
8. Alice computer secret key $= (B^a \bmod P)$
9. Bob computer secret key $= (A^b \bmod P)$
10. Both get same shared secret key.

Output:

Input: $P = 23$
$g = 5$
Alice private key = 6
Bob private key = 15

Output:

Alice public key = 8
Bob public key = 19
Shared secret key = 2 = 2.

8. Convert plain text to RSA algorithm

Aim:

To implement RSA algorithm for some encryption and decryption of a message using public and private key.

Algorithm:

1. Enter two prime numbers p and q.
2. Compute $n = p * q$
3. Compute $\phi(n) = (p-1)(q-1)$
4. Choose such that $\gcd(e, \phi) = 1$
5. Compute & sum that $(d \times e) \bmod \phi = 1$
6. Public key $= (e, n)$
7. Private key $= (d, n)$
8. Encrypt message using $c = (m^e \bmod n)$
9. Decrypt message using $M = (C^d \bmod n)$

Output:

Input:

$p = 61$
$q = 53$
$e = 17$
plain text = HI

output:

public key = (17, 3233)
private key = (2753, 3233)
Encrypted = (3000, 1313)
Decrypted = HI

9 TO Convert plain text to triple DES algorithm

Aim:

TO implement triple DES algorithm python to secure data transmission.

Algorithm:

1. start the program.
2. Import required libraries for triple DES.
3. Display menu for encryption or decryption.
4. Read the User Choice and input message.
5. Generate a 24 bytes secret key.
6. Create Cipher object using the key.
7. Encrypt the plaintext to produce cipher text.
8. Decrypt the cipher text to obtain original plain text.
9. Display the output.

output:

Input

plain text : NETWORK

key : 123456789012345678901234

output:

encrypted : 5F4DCC3B5AA765D6

Decrypted : NETWORK

10. To Convert plain text to Blow fish algorithm

Aim:

To implement the blowfish encryption algorith to securely encryt and decrypt data using a symmetric key block cipher.

Algorithm:

1. Start the program.
2. Read the plaintext and Secret key.
3. Initialize P-array and S-boxes.
4. Divide the plaintext into two halves.
5. perform 16 rounds of Feistel operations using the key.
6. Combine the two halves to get the Cipher text
7. perform reverse steps for decryption.
8. stop the program.

Output:

Input: 8 bit binary text: 10101010
binary key: 1010

Output:

Encrypted: 11110101

11. To Convert plain text to RC5 algorithm

Aim:

To implement the RC5 algorithm to encrypt and decrypt data using variable key size and rounds.

Algorithm:

1. Start the program.
2. Read the plaintext and Secret Key.
3. Initialize parameters (word size, rounds, key, length).
4. Generate subkeys using key expansion.
5. Divide plaintext into two blocks (A and B)
6. perform encryption rounds using XOR and rotation operations.
7. Combine blocks to produce ciphertext.
8. perform reverse steps for decryption.
9. Stop the program.

Output:

1. Encrypt 2. decrypt

A: 10

B: 20

Key: 5

Round: 3

Encrypted : 250 | 18